INCH-POUND
MIL-DTL-24643/78
27 November 2023

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, -4 DEGREES FAHRENHEIT (-20 DEGREES CELSIUS) TO +176 DEGREES FAHRENHEIT (+80 DEGREES CELSIUS), 300 VOLTS, TYPE LSCBPSJ (INCLUDING VARIANT LSCBPSJW)

This specification is approved for all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24643.

This cable is one or two twisted pair(s) with overall shield, intended for use with CAN-Bus data communication networks. This cable can be manufactured in two variations: LSCBPSJ (non-watertight) and LSCBPSJW (watertight).

Construction (Non-Watertight and Watertight)

First - Copper conductor, tin-coated, in accordance with ASTM B286 (see table I for sizes).

Second - Polyolefin insulation (see table I for insulated conductor finished diameter).

Third - Two insulated conductors twisted together with a lay length of 20 to 32 times the maximum specified diameter of a single insulated conductor.

Wire identification shall be as follows:

Pair 1 – Green and Yellow
Pair 2 – Brown and White

NOTE: This step is not applicable to cables with only one pair in them.

Fourth - Two singles (one Green and one Yellow, if one pair cable), or two pairs (see table II or table III for number of components required), shall be cabled together with a lay in accordance with 3.3.4 of MIL-DTL-24643. Fillers may be used as necessary to form a firm, well-rounded assembly.

Fifth - An optional binder.

Sixth - Metallic foil/polyester tape. Metallic side facing out and in contact with the braid. Tape may be longitudinally or helically applied, with minimum overlap of 10 percent.

Seventh - Braided shield of tin-coated copper.

Eighth - An optional binder.

Ninth - Cross-linked polyolefin jacket. Jacket wall thickness shall be 0.030±0.005 inch for cable with a core diameter less than 0.300 inch and shall be 0.040±0.005 inch for cable with a core diameter greater than or equal to 0.300 inch.

Distribution Statement A. Approved for public release: distribution is unlimited.

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TABLE I. Component wire details.

Wire size (AWG)	Stranding (no. x AWG)	Conductor diameter (inch ±0.003)	Insulated conductor finished diameter (inch)	
			min.	max.
22	7 x 30	0.030	0.079	0.085
20	7 x 28	0.038	0.097	0.103
18	19 x 30	0.049	0.066	0.072

TABLE II. Details (non-watertight).

Military part no. M24643/78	Type and size	Wire size (AWG)	No. of pairs	Conductor resistance (ohms)	Overall diameter (inch)	
					min.	max.
-01UO	LSCB1PSJ-22	22	1	5.64	0.237	0.257
-02UO	LSCB2PSJ-22	22	2	5.75	0.336	0.356
-03UO	LSCB1PSJ-20	20	1	3.51	0.273	0.293
-04UO	LSCB2PSJ-20	20	2	3.58	0.410	0.430
-05UO	LSCB1PSJ-18	18	1	2.08	0.331	0.351
-06UO	LSCB2PSJ-18	18	2	2.12	0.506	0.526

TABLE III. Details (watertight).

Military part no. M24643/78	Type and size	Wire size (AWG)	No. of pairs	Conductor resistance (ohms)	Overall diameter (inch)	
					min.	max.
-07UO	LSCB1PSJW-22	22	1	5.64	0.243	0.273
-08UO	LSCB2PSJW-22	22	2	5.75	0.362	0.392
-09UO	LSCB1PSJW-20	20	1	3.51	0.279	0.309
-10UO	LSCB2PSJW-20	20	2	3.58	0.419	0.449
-11UO	LSCB1PSJW-18	18	1	2.08	0.337	0.367
-12UO	LSCB2PSJW-18	18	2	2.12	0.512	0.542

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REQUIREMENTS:

Qualification required.

INSPECTION:

Basic electricals:

Conductor resistance (ohms/100 meters at 77 °F [25 °C], max.)	See table II or table III
Conductor resistance unbalance (percent, max.)	5
Voltage withstand	
Conductor to conductor	1.0 kVAC or 2.0 kVDC
Conductor to overall shield	1.0 kVAC or 2.0 kVDC
Insulation resistance (megohms/100 meters, max.)	
Conductor to conductor	Not applicable
Conductor to overall shield	Not applicable
Conductor and shield continuity	No failure
Jacket flaws	No failure

Group A:

Visual and dimensional	No failure
Watertightness (type LSCBPSJW only) (see MIL-DTL-24643 for limits of water leakage)	No failure
Capacitance	
Mutual capacitance (nominal at 1 kilohertz [kHz]) (nanofarad [nF]/100 meters)	75
Between one conductor and other conductor(s)	
Connected to shield, at 1 kHz (picofarad [pF]/100 meters, max.)	110
Characteristic impedance, at 1 megahertz (ohms)	120±12

Group B:

Cold bending (cable)	No failure
Drip (203±1.8 °F [95±1 °C]) (type LSCBPSJW only)	No failure
Physicals (unaged)	
Insulation (extruded insulation only)	
Tensile strength (lb/in^2, min.)	700

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Elongation (percent, min.)	150
Jacket (cable)	
Tensile strength (lb/in^2, min.)	1,300
Elongation (percent, min.)	160
Tear (lb/in thickness, min.)	35
Thermoset proof test (percent, max.)	
Insulation	Not applicable
Jacket (when tested at 392 °F [200 °C])	50

Group C:

Physicals (aged) (air oven)	
Insulation (extruded insulation only)	
Tensile strength (percent of unaged, min.)	75
Elongation (percent of unaged, min.)	75
Jacket (cable)	
Tensile strength (percent unaged, min.)	60
Elongation (percent of unaged, min.)	60
Permanence of printing (jacket) (cycles, min.)	125
Heat distortion (percent of unaged, max.)	30
Cable fill (sealant) removability (type LSCBPSJW only)	No failure
Shrinkage	No failure
Surface transfer impedance	
Milliohms per meter (max.)	700
Electromagnetic pulse (EMP) response (dB, min.)	60

Group D:

Flame propagation (cable)	No failure

Qualification inspection:

Qualification inspection shall include basic electricals; Groups A, B, C, and D; plus the following:

Stress resistance	Not applicable
Temperature humidity	No failure
Aging and compatibility (cable) (203±3.6 °F [95±2 °C])	No failure
Abrasion resistance (jacket) (scrapes, min.)	75

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Acid gas equivalent (percent, max.)	
Jacket	2
Fillers	2
Insulation	18
Halogen content (percent, max.)	
Jacket	0.2
Fillers	0.2
Insulation	0.2
Immersion (jacket)	
Tensile strength (percent of unaged, min.)	50
Elongation (percent of unaged, min.)	50
Smoke index (max.)	
Jacket	25
Fillers	45
Insulation	45
Toxicity index (max.)	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) – type A (hardness, min.)	80
Weathering (jacket)	No failure
Electrical moisture absorption	No failure

UNIT ORDERING LENGTHS:

Type and size	Feet (nominal)
All types and sizes	500

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CONCLUDING MATERIAL

Custodians:
Army – MI
Navy – SH

Preparing activity:
Navy – SH
(Project 6145-2017-008)

Review activities:
Army – AR, CR
Navy – CG, EC
DLA – CC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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